

Trading Rules

Sports betting operates differently from equity or futures trading, so terms like "buy" and "sell" need to be adapted. In this context, "opening a position" means placing a bet, and "closing a position" means the bet settling at the final whistle (or cashing out early on a betting exchange). There is no continuous holding period like a stock position. The bet is placed pre-match and resolves when the match ends.

Rule 1: Signal Generation

For every match in scope (World Cup, major domestic leagues, continental tournaments), run the fine-tuned LLM with a structured prompt containing both teams' full rosters and per-player statistics. The model outputs three things: predicted probabilities for each outcome (home win, draw, away win), a predicted scoreline, and a reasoning chain. This step happens 24-48 hours before kickoff to allow time for human review.

Rule 2: When to Open a Position (When to "Buy")

A position is opened when all three conditions are met:

Condition A (Quantitative Edge): The model's predicted probability for at least one outcome exceeds the bookmaker's implied probability by more than the edge threshold T_{edge} . For example, if the bookmaker's odds imply a 40% chance of a draw, and the model predicts 58%, the edge is 18 percentage points. If T_{edge} is set to 10%, this qualifies.

Condition B (Reasoning Audit): A human reads the model's reasoning chain and confirms that the logic is factually sound. Specifically: all players cited in the reasoning are available (not injured, suspended, or dropped from the squad), the statistics referenced are current (not from two seasons ago), and the causal logic is coherent (the reasoning connects player attributes to match dynamics in a way that makes football sense). If the reasoning fails this check, the signal is discarded regardless of the quantitative edge.

Condition C (Risk Budget): The bet does not cause total open exposure to exceed the portfolio-level cap (see Rule 4). If placing this bet would push total exposure above the limit, it is skipped.

If any of these three conditions is not met, the match is skipped entirely. No partial bets, no "let me bet a small amount anyway." Discipline is the whole point of a rule-based system.

Rule 3: Position Sizing (How Much to Bet)

Position size is determined by a fractional Kelly criterion: $\text{Bet size} = f * \text{Kelly optimal bet}$

Where the full Kelly optimal bet is: $\text{Kelly \%} = (p * (d - 1) - (1 - p)) / (d - 1)$

Here, p is the model's predicted probability of the chosen outcome, d is the decimal odds offered by the bookmaker, and f is the Kelly fraction (a conservatism parameter, set between 0.1 and 0.5 of full Kelly).

Full Kelly is mathematically optimal for long-run bankroll growth but is extremely aggressive and can produce large drawdowns. Fractional Kelly (using $f = 0.25$ as a starting point) sacrifices some expected growth for significantly smoother equity curves. The exact value of f is a tunable parameter that should be calibrated during backtesting.

Additionally, the bet size is hard-capped at $E_{\max}$ percent of the current bankroll per match (starting value: 3%). Even if Kelly suggests a larger position, the cap is enforced. This prevents any single match from causing catastrophic loss.

Rule 4: Portfolio-Level Risk Management

Maximum concurrent exposure: Total wagered across all unresolved bets must not exceed 15% of bankroll at any time. On a typical matchday with multiple games, this forces prioritization. Bets are ranked by edge magnitude and placed in descending order until the exposure cap is reached.

Drawdown circuit breaker: If cumulative realized losses reach D_{stop} percent of the starting bankroll for the current evaluation period (e.g., a tournament or a league season), all betting activity is paused. The model and reasoning outputs are reviewed for systematic errors before resuming. Starting value for D_{stop} : 20%.

Consecutive loss limit: If 8 consecutive bets lose, pause and review regardless of whether the drawdown threshold has been reached. This catches situations where the model may be systematically miscalibrated on a particular league or match type.

Rule 5: When a Position Closes (When to "Sell")

The default is that positions close at the final whistle. The bet settles based on the actual match result, and the profit or loss is recorded.

On betting exchanges (like Betfair) that allow in-play trading, there is one exception: if material information emerges between bet placement and kickoff (e.g., a key player is suddenly ruled out with an injury after the bet was placed, and that player was central to the model's reasoning), the position may be closed early by laying off the bet at current exchange odds. This is the only scenario where early exit is permitted. We do not trade in-play based on live match events.

Rule 6: Record Keeping and Feedback Loop

Every bet is logged with: match details, model's predicted probabilities and reasoning, bookmaker odds at time of placement, bet size, actual outcome, and profit/loss. After each evaluation period, the following are reviewed:

Prediction accuracy: What percentage of bets were correct? Is accuracy above baseline (random weighted by historical outcome distribution)?

Reasoning quality: For losing bets, was the reasoning logically sound but the outcome was just variance? Or did the reasoning contain identifiable flaws (overweighting a metric, ignoring defensive matchups, etc.)? This distinction matters because a good reasoning process that produces a losing bet is fine, but a flawed reasoning process that happens to produce a winning bet is dangerous.

Calibration: When the model predicts 70% probability, does that outcome occur roughly 70% of the time across all bets? If the model is systematically overconfident or underconfident, the edge threshold T_{edge} should be adjusted accordingly.

These logs feed directly into the next round of model fine-tuning. Matches where the model's reasoning was clearly wrong become high-priority training samples for the next iteration.

Summary of Parameter Starting Values

Parameter	Definition	Starting Value	Range
T_edge	Minimum edge to trigger a bet	10%	5-20%
f	Kelly fraction for position sizing	0.25	0.1-0.5
E_max	Max bet size per match (% of bankroll)	3%	1-5%
Exposure_max	Max total open exposure	15%	10-25%
D_stop	Drawdown circuit breaker	20%	10-25%
Consecutive loss limit	Pause trigger	8 losses	5-12